Illumination Invariant Feature Based on Neighboring Radiance Ratio

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***Abstract*—In many object recognition applications, especially in face recognition, varying illuminations can adversely affect the robustness of the object recognition system. In this paper, we propose a novel illumination invariant feature called Neighboring Radiance Ratio (NRR) which is insensitive to both intensity and direction of light. NRR is derived and analyzed based on a physical image formation model. The computation of NRR does not need any prior information or any training data and NRR is far less sensitive to the border of shadows than most existing methods. The analysis of the illumination invariance of NRR is also presented. The proposed NRR feature is tested on Extended Yale B and CMU-PIE databases and compared with several previous methods. The experimental results corroborate our analysis and demonstrate that NRR is highly robust image feature against illumination changes.**

***Index Terms*—Neighboring Radiance Ratio, Illumination invariant, object recognition.**

I. Introduction

For most object recognition tasks, a challenge is to maintain recognition accuracy and consistency against wide variations of illumination in input images. In face recognition, for instance, images of the same person taken in varied illumination conditions may appear drastically different, due to shadows cast by the 3D structures of the human face. Variations in images of the same person due to varied illuminations may be greater than the variations in images of different persons. The illumination-induced variation has been one of the main obstacles in accurate and robust face recognition.

A number of methods have been proposed to factor out illumination variations in object recognition. Classical methods such as linear discriminant analysis (LDA) [1] and principal component analysis (PCA) [2] could reduce the image into a low dimensional linear subspace. Georghiades [3] proposed to build a 3D linear subspace by using three or more images from same person under varying illumination conditions. Basri and Jacobs [4] utilize a 9-dimensional linear subspace to model the intensity of the same surface under different illumination. Lee et al. [5] suggest a subspace by nine images with illumination from different nine directions.

Other methods like image normalization methods and preprocessing methods also achieve good results to some degree without using illumination models. One of the most well-known methods is Histogram Equalization (HE) technologies. Xie and Lam [6] proposed an efficient illumination normalization method in which the intensity of pixels is locally normalized using a surrounding window. Chen et al. [7] proposed a discrete cosine transform (DCT) where illumination variance mainly depends on the low-frequency subbands.

Illumination invariant feature extraction is another category of techniques to improve recognition robustness against illumination variations. Local binary pattern (LBP) [8] is a well-known illumination invariant method which thresholds the 3x3-neighborhood of each pixel with the center value and generate a binary number. Gao and Leung proposed line edge map in [9] to extract illumination invariant features. Shashua and Riklin-Raviv [10] demonstrated Quotient Image method (QI) determines the ratio between a given image and linear combination of other three images under independent illumination conditions. After that, Wang et al. [11] proposed Self Quotient Image which is actually a multiscale retinex approach and easier than Quotient Image method as it does not need any training data. Nabatchian et al. [12] proposed a simple but effecteive method based on retinex model to extract illumination invariant feature by using maximum filter to calculate the low frequency part of the image. Recently, Baradarani et al. [13] proposed to utilize double-density dual-tree complex wavelet transform (DD-DTℂWT) to remove illumination in object images.

In this paper, we propose a novel illumination invariant feature called Neighboring Radiance Ratio (NRR) which is insensitive to both intensity and direction of light. The analysis of invariance to both light intensity and direction of NNR is also presented. We propose that the ratio of adjacent pixels' value should be illumination invariant in both intensity and direction. Which is superior than most previous methods, NRR is far less sensitive to the border of shadows and needs no any prior knowledge and training data. To verify the effectiveness and superiority of our approach, we perform experiments on Extended Yale B [14] and CMU-PIE face database [15] and compare results with several previous methods in the literature. The rest of paper is organized as follows: Neighboring Radiance Ratio is introduced in Section 2. The analysis of invariance to light direction of NRR will be presented in

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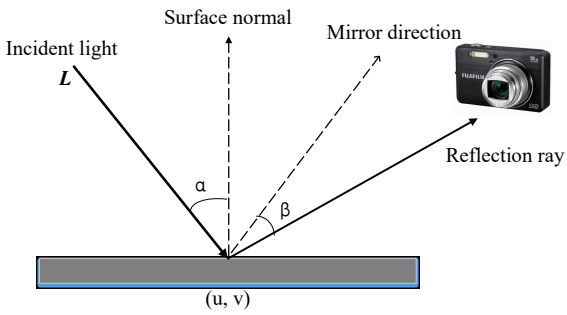


Fig. 1. The illustration of the physical image formation model. In the diagram, α is the angle between incident light and object surface normal at pixel (u, v), β is the angle between reflection ray and mirror direction of the incident light.

Section 3. Experiments and results are shown in Section 4 and finally the conclusion is presented in Section 5.

II. Neighboring Radiance Ratio

A. Background

In quest for illumination invariant features, it is necessary to examine how the intensity of a surface point relates to the strength and direction of the incident light impinging on that point. We consider the physical image formation model:

$$I(u,v) = (cos\alpha + \eta cos^n \beta) \int_{\Lambda} \gamma(\lambda) L(\lambda) \rho_{u,v}(\lambda) d\lambda \qquad (1)$$

where (u, v) is the pixel location on an object surface, α is the angle between incident light and object surface normal, β is the angle between reflection ray and mirror direction of the incident light, $\eta \in [0, 1)$ is specular coefficients and $n > 1$ is the degree of surface shininess. In addition, $\gamma(\lambda)$ is camera's response to the k-th band of reflection ray, $L(\lambda)$ is the spectrum of incident light and $\rho_{u,v}(\lambda)$ is the reflectance of pixel (u, v) in object surface. The schematic diagram of physical image formation model is shown in Figure 1.

B. Neighboring Radiance Ratio

In the image formation model, the spectral response $\gamma(\lambda)$ is fixed for a given camera. The spectrum of the light source $L(\lambda)$ can also be considered the same in a neighborhood area. As the reflectance $\rho_{u,v}(\lambda)$ is determined by the surface material, it is also a constant in a surface area consisting of a uniform material (e.g., skin or hair). If two adjacent pixels I_i and I_j are imaged of the same surface material, then we have $\rho_i = \rho_j$ and consequently,

$$\frac{I_i}{I_j} = \frac{cos\alpha_i + \eta cos^n \beta_i}{cos\alpha_j + \eta cos^n \beta_j} \qquad (2)$$

Fixed the direction of the impinging light L, the ratio I_i/I_j, which is called the Neighboring Radiance Ratio (NRR) in sequel, is a function of the two angles α and β but independent of the strength of the light source L. In the next section we will show that NRR is also insensitive to the direction of the light source. For these reasons the NRR is by nature an illumination invariant feature that carries the information on the surface geometry and material of a local area.

However, the definition of NRR needs to be refined in the presence of shadows. The illumination invariance of NRR holds only if both pixels I_i and I_j in (2) are in or out of the shadow. To meet this requirement, for a given pixel I_i we choose its neighboring pixel I_j in NRR carefully as follows.

The first temptation is to choose the neighboring pixel whose value is the closest to I_i as the I_j in NRR. However, the NRR feature has to be not only illumination invariant but also highly sensitive to surface properties in order to be useful in recognition and classification applications. For these reasons we need to choose I_j to be the neighbor of I_i that differs the most in value from I_i while being on the same side of the shadow (if it exists) as I_i. To achieve this we find the minimum pixel value m_i and the maximum pixel value M_i in the 3×3 neighborhood of I_i. If pixel I_i is at shadow boundary, then the two pixels corresponding to m_i and M_i must be on the different sides of the shadow boundary. For making I_j and I_i to be at the same side of the shadow boundary, we choose

$$I_j = \begin{cases} m_i & if \quad |I_i - m_i| \leq |I_i - M_i| \\ M_i & \text{otherwise} \end{cases} \qquad (3)$$

.

After normalizing the value of NRR to the range $[0, 1]$, we arrive at the final definition of the NRR feature at pixel i:

$$\tau = \frac{min\{I_i, I_j\}}{max\{I_i, I_j\}} \qquad (4)$$

III. Invariance to light direction

In this section, we show that the feature NRR is not only independent of the strength of the illumination, it is also insensitive to the direction of the light. To simplify analysis, we only consider the Lambertian reflection. This approximation is acceptable because the dominant portion of the total reflection of most object surfaces is Lambertian.

Considering that the direction of incident light varies in a range $\alpha \in [-\theta, \theta]$, we measure how much the Neighboring Radiance Ratio τ can accordingly change. In order to keep the value of NRR in the range $[0, 1]$, we need to account for possible reversal of the relative order of the values of two neighboring pixels, and rewrite equation (4) as the following piecewise function:

$$\tau = \begin{cases} \frac{cos(\alpha+\epsilon)}{cos\alpha} & \alpha \in [-\frac{\epsilon}{2}, \theta] \\ \\ \frac{cos\alpha}{cos(\alpha+\epsilon)} & \alpha \in [-\theta, -\frac{\epsilon}{2}) \end{cases} \qquad (5)$$

where $\epsilon > 0$ is the change of the surface normal between adjacent pixels i and j ($\alpha_j = \alpha_i + \epsilon$), as depicted in Figure 2.

To evaluate the rate of change of NRR τ in variable α, we differentiate τ with respect to α:

$$\frac{\partial \tau}{\partial \alpha} = \begin{cases} \frac{-sin\epsilon}{cos^2\alpha} & \alpha \in [-\frac{\epsilon}{2}, \theta] \\ \\ \frac{sin\epsilon}{cos^2(\alpha+\epsilon)} & \alpha \in [-\theta, -\frac{\epsilon}{2}) \end{cases} \qquad (6)$$

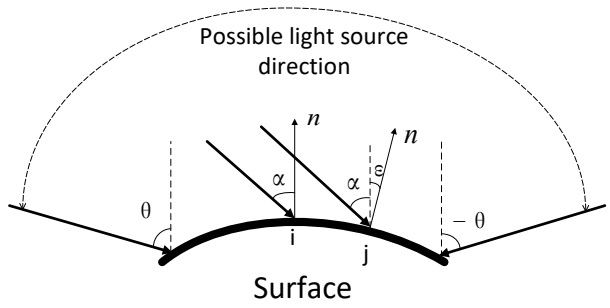


Fig. 2. Illustration of surface normals $\vec{n}$, light direction α, and the range of changing light directions $[-\theta, \theta]$.

As the change of the surface normal ϵ is very small between two adjacent pixels for smooth surfaces, the partial derivative $\partial\tau/\partial\alpha$ is also very small, as shown in Figure 3(a); namely, NRR τ remains almost a constant over $\alpha \in [-\theta, \theta]$.

It follows from equation (6) that the maximum and minimum value of τ in $[-\theta, \theta]$ are respectively:

$$\tau_{max} = 1, \quad \tau_{min} = \frac{cos(\theta + \epsilon)}{cos\theta} \tag{7}$$

so the maximum change of the ratio τ is

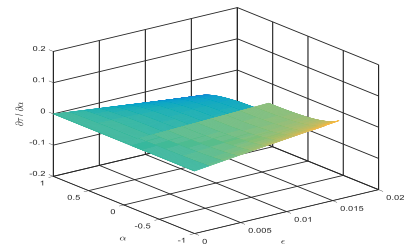
$$|\Delta\tau|_{max} = 1 - \frac{cos(\theta + \epsilon)}{cos\theta}, \tag{8}$$

when the angle of incoming light sweeps from $-\theta$ to θ. Figure 3(b) plots $|\Delta\tau|_{max}$ as a function of θ and ϵ and it shows stability of the NRR feature against changing light directions.

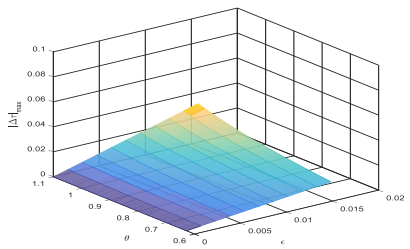
IV. Experimental Results

The above analyses on the robustness of the proposed illumination invariant NRR feature are corroborated by our experimental results reported in this section. The effectiveness of the new NRR feature is evaluated in a case study of face recognition. The extended-Yale B and CMU-PIE face databases are tested in our experiments because they provide face images acquired in extreme illumination conditions (see Figure 4). In data preparation the face images in Extended Yale B database are all aligned and cropped to 192x168 resolution; the face images in CMU-PIE database are all cropped to 64x64 resolution.

The extended Yale B Face Database contains 16128 images of 28 human subjects under 9 poses and 64 illumination conditions. Our experiments are only concerned with face images in frontal pose. We divide the database into five different subsets by the angle between the camera axis and the light direction. This camera-light angle described by two parameters, azimuth (A) and elevation (E). The above five subsets of the Extended Yale B database are: Subset 1 consisting of images of $max\{A, E\} \leq 10^o$, Subset 2 consisting of images of $10^o < max\{A, E\} \leq 25^o$, Subset 3 consisting of images for $25^o < max\{A, E\} \leq 50^o$, Subset 4 consisting of images for $50^o < max\{A, E\} \leq 75^o$, Subset 5 consisting of images of $max\{A, E\} > 75^o$. For the Extended YaleB



(a)



(b)

Fig. 3. (a) The partial derivative $\partial\tau/\partial\alpha$ as a function of α and ϵ; (b) The maximum change $|\Delta\tau|_{max}$ as a function of θ and ϵ.

database, we use images in Subset 1, which are taken under relatively ideal light source for training and the other images in the four other subsets for testing.

The CMU-PIE database contains 41368 images from 68 subjects under 43 different illumination conditions and 13 different poses. Same as above, we just use the frontal view images. For each subject, there are 43 frontal view images under different illumination conditions, 22 images of them are taken under light-on conditions and the other 21 images are taken under light-off conditions. For CMU-PIE database, we perform experiments on light-on set, light-off set and the union of the two sets (named light-on/off set) respectively. In light-on set, for each subject, we randomly select 4 images for training and the other images (18 images) for testing. In light-off set, we randomly select 3 images for training and the other images (18 images) for testing. In light-on/off set, we select 5, 6, 7 images respectively for training and the other images for testing.

We evaluate the effectiveness of the NRR image feature by coupling it with the KNN and SVM algorithms, two of the well-known feature-based classifiers. The modular PCA [16] is performed on the NRR images to extract features for them.

Figure 4 shows the NRR images of some samples in the Extended Yale B database. Visually, one can see that the NRR images achieve a good degree of consistency over a wide range of different illumination conditions.

The recognition accuracy results are tabulated in Tables I and II. For the Extended Yale B database, the recognition rate with NRR is higher than other methods in most cases. Only in Subset 5, the recognition accuracy with NRR is slightly lower

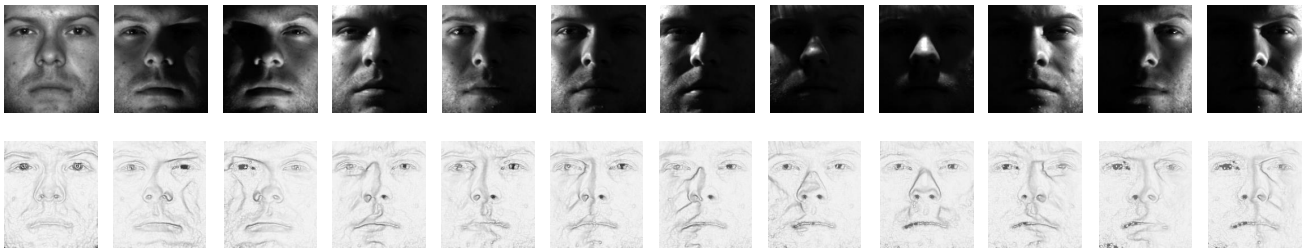


Fig. 4. Neighboring Radiance Ratio images of some samples in the Extended Yale B database. Top row: face images in the Extended Yale B database; bottom row: the corresponding Neighboring Radiance Ratio images.

TABLE I
RECOGNITION RESULTS ON EXTENDED YALE B DATABASE.

Method	Subset 2	Subset 3	Subset 4	Subset 5
PCA	90.16%	41.23%	6.37%	3.24%
LDA	100%	98.12%	38.35%	5.13%
II+PCA+SVM	100%	99.78%	95.44%	94.68%
II+PCA+KNN	100%	100%	96.01%	92.44%
II+PCA+DVS	100%	100%	96.60%	95.40%
II+PCS+WVS	100%	100%	97.91%	96.54%
DD-DT$\mathbb{C}$WT	100%	100%	96.68%	99.03%
NRR+PCA+KNN	100%	100%	97.81%	97.53%
NRR+PCA+SVM	100%	100%	98.90%	98.34%

TABLE II
RECOGNITION RESULTS ON CMU-PIE DATABASE.

light	training ratio	NRR	DD-DT$\mathbb{C}$WT	II
On	4/22	100%	100%	99.99%
Off	3/21	99.60%	99.59%	94.85%
On/Off	5/43	99.83%	99.70%	98.99%
On/Off	6/43	99.90%	99.77%	99.55%
On/Off	7/43	99.93%	99.92%	99.74%

than DD-DT$\mathbb{C}$WT, but NRR achieves higher recognition rate in Subset 4 and it is computationally more efficient than DD-DT$\mathbb{C}$WT. As shown in Table II, the recognition accuracy with NRR is uniformly higher than DD-DT$\mathbb{C}$WT [13] and II [12] in the experiments on the CMU-PIE database.

The experimental results above corroborate our analyses in Sections 2 and 3, establishing the robustness of the proposed illumination invariant NRR image feature against greatly varied illumination conditions at the stage of image acquisition.

V. CONCLUSION

In this paper, we propose a new robust image feature called Neighboring Radiance Ratio, and analytically establish the illuminance invariance of NRR. Our analyses are corroborated by an empirical study of using the NRR image feature for face recognition task. Experimental results show that the NRR image feature can withstand large changes in the strength and direction of the illumination in acquired images and still achieve superior recognition performances under extreme lighting conditions.

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